RKDF University

Program Outcomes, Program Specific Outcomes and Course Outcomes

1. Faculty of Agriculture

B.Sc (Hons) Agriculture

Program Outcomes

PO: 01 To impart firsthand knowledge on agriculture and allied sciences

PO: 02 To impart in-depth practical knowledge in agriculture and allied sciences

PO: 03 To provide extensive knowledge on agri-allied sectors like livestock, Poultry

PO: 04 To disseminate different technologies through various extension activities

PO: 05 To identify and overcome the problems encountered in day-to-day agriculture

PO: 06 To provide knowledge on commercial agricultural production practices

PO: 07 To make students competitive in pursuing higher studies

Program Specific Outcomes

PSO:01 Agriculture knowledge: Apply the knowledge of horticulture, Agronomy, organic and sustainable agriculture, and integrating pest management to the solution of Agriculture related issues.

PSO:02 Analysis of complex problems: To understand and analyze the current events and issues that are occurring in agriculture and how they affect futuristic agriculture. Able to demonstrate critical thinking and problem solving skills as they apply to a variety of animal and or plant production systems.To understand problem solving skills in crop production and animal husbandry.

PSO:03 Modern tool usage: Select and apply advanced techniques, resources and IT tools for prediction weather. Knowledge of Weather codes and Symbols, Reading and

Recording of weather and climatic data.To get trained for climatologically records, soil data and soil nutrition.

PSO:04 The Agriculturist and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional Agricultural practice

PSO:05 Environment and sustainability: Understand the impact of the professional agricultural solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PSO:06 Ethics: To demonstrate research based knowledge of the legal and ethical environment impacting agriculture organizations and exhibit an understanding and appreciation of the ethical implications of decisions. In accordance with high standards of academic integrity (ethics and moral) both in the profession and in society as a whole.To develop competence to work in Government, public and private sectors.

PSO:07 Individual and team work: To demonstrate an ability to engage in critical thinking by analyzing situations and constructing and selecting viable solutions to solve problems. Abilities to work effectively with each other. To develops analytical ability and team work spirit.

PSO:08 Communication: To demonstrate and understand the impact of globalization and diversity in modern agriculture organizations. Understanding working of SHG, NGO, Govt Extension service agencies. Communicate effectively with the farmer community and with society at large. Be able to comprehend and write effective reports documentation. Make effective presentations, and give and receive clear instructions.

PSO: 09 Project management and finance: Demonstrate knowledge and understanding of Agriculture and Agri business management principles. Understand how all aspects of agriculture combine and are used by scientists, marketers, producers and understand how employer characteristics and decision-making at various levels enhance the success of an agricultural enterprise. To understand components of agri business and economics of market.

PSO:10 Entrepreneurship and employability: Able to recognize and examine the relationships between inputs and outputs in their agricultural field to make effective and profitable decisions. To understand the mechanics of agripreneurship. This programme will also help students to enhance their employability for jobs in different sectors.

PSO:11 Subject specific knowledge: Demonstrate knowledge and understanding in horticulture section: The breadth and depth of the profession of horticulture. Basic horticulture biology: taxonomy, anatomy, morphology, and physiology. The characteristics of the environment and their influence on plant growth and development, Current applications of horticultural principles and practices: propagation, pest management, production, maintenance, and business practices. Provide comprehensive knowledge of horticultural production.

PSO: 12 Life-long learning: To develop critical and self-critical opinion and approach aiming at solving the most important practical problems in the field of sustainable agriculture by applying gained competencies. The graduates will generate a culture of lifelong learning in an inclined environment to achieve personal enrichment and professional ethics.

Course Outcomes

BAG-5101	**Fundamentals of Horticulture**	**CO1**	To get the knowledge about diversity of Plants and identification of plant vegetative
		CO2	Students will be able to identify plant vegetative
		CO3	Information to solve horticultural problems
		CO4	Information about different types of irrigation on different plants.
		CO5	To understand about fertilizers application methods
BAG-5102	**Fundamentals of Plant Biochemistry and Biotechnology**	**CO1**	To understand the importance of pH
		CO2	Information about different nutrients and their importance.
		CO3	Importance of DNA and RNA (Nucleic acids)
		CO4	To understand the role of metabolism
		CO5	Role of enzymes in food industry.
BAG-5103	**Fundamentals of Soil Science**	**CO1**	Students will gain knowledge on concepts and principles of Soil Science
		CO2	Relevant knowledge on rocks and minerals, their composition and the types of soils formed from different parent materials.
		CO3	Understand the role of soil forming factors and processes in soil formation
		CO4	Understand various soil physical, chemical and biological properties and their impact on plant growth.
		CO5	The knowledge gained in this course will be useful in understanding the role of soils in crop production and management.
BAG-5104	**Introduction to Forestry**	**CO1**	Information about the provision of timber
		CO2	To gets the knowledge about Fuel wood
		CO3	To study about wildlife habitat
		CO4	Natural water quality management
		CO5	Recreation
BAG-5105	**Comprehension & Communication Skills in English**	**CO1**	Demonstrate a significant increase in word knowledge.
		CO2	Employ prereading, skimming, and prewriting techniques.
		CO3	Identify main ideas in paragraphs and reading selections.
		CO4	Locate important details.
		CO5	Decipher paragraph patterns, writer techniques, and conclusions.
BAG-5106	**Fundamentals of Agronomy**	**CO1**	In modern terminology however the word has come to mean and denote a branch of science dealing with all aspects of crop cultivation and production.
		CO2	A study of agronomy often involves a summoning of resources from related disciplines such as Botany, Soil Science, Irrigation, plant protection, Plant Genetics and Breeding, Agro-meteorology etc.
		CO3	In a more fundamental sense it can be categorized as an applied Science, the object of which is crop cultivation and management for the purpose of producing food for humans, feed for animals as well as raw materials for the industry.
		CO4	Knowledge about Indian Agriculture and importance, present status, scope and future prospect
		CO5	Cropping seasons of India. Soil formation, classification, physical, chemical properties. Identification of important crops and crop seeds.
BAG-5107B	**Introductory Biology***	**CO1**	Understand critically interpret the primary biological literature in his/her area of interest.
		CO2	Understand design, conduct, analyze, and

			communicate (in writing and orally) biological research.
		CO3	Understand basic ethical principles to basic and applied biological/biomedical practice and will understand the role of biological/biomedical science, scientists, and practitioners in society.
		CO4	Understand organic evolution and its underlying principles and mechanisms.
		CO5	Understand fundamental biological processes of metabolism, homeostasis, reproduction, development, and genetics, and the relationships between form and function of biological structures at the molecular, cellular, organismal, population, and ecosystem levels of the biological hierarchy.
BAG-5107A	**Elementary Mathematics***	**CO1**	Demonstrate competency in the areas that comprise the core of the mathematics major
		CO2	Demonstrate the ability to understand and write mathematical proofs
		CO3	Be able to use appropriate technologies to solve mathematical problems
		CO4	Be able to construct appropriate mathematical models to solve a variety of practical problems
		CO5	Obtain a full-time position in a related field or placement
BAG-5108	**Agricultural Heritage**	**CO1**	Demonstrate competency in the areas that comprise the core of the mathematics major
		CO2	Demonstrate the ability to understand and write mathematical proofs
		CO3	Be able to use appropriate technologies to solve mathematical problems
		CO4	Be able to construct appropriate mathematical models to solve a variety of practical problems
		CO5	Obtain a full-time position in a related field or placement
BAG-5109	**Rural Sociology & Educational Psychology**	**CO1**	Understand concept of rural sociology, its importance in agricultural extension, characteristics of Indian rural society.
		CO2	Understand social groups, social stratification, culture, social values, social control and attitudes, leadership and training.
		CO3	Understand concept of educational psychology, intelligence, personality, perceptions, emotions, frustration, motivation, teaching and learning
		CO4	Acquaint with characteristics of rural society, village institutions and social organizations. Select lay leaders and train them.
		CO5	Assess personality types, leadership types and emotions of human beings
BAG-5110	**Human Values & Ethics**	**CO1**	Understand the significance of value inputs in a classroom and start applying them in their life and profession.
		CO2	Distinguish between values and skills, happiness and accumulation of physical facilities, the Self and the Body, Intention and Competence of an individual, etc.
		CO3	Understand the value of harmonious relationship based on trust and respect in their lifeand profession.
		CO4	Understand the role of a human being in ensuring harmony in society and nature.
		CO5	Distinguish between ethical and unethical practices, and start working out the strategy to actualize a

			harmonious environment wherever they work.
BAG-5111	**NSS/NCC/Physical Education & Yoga Practices**	**CO1**	Learn to work in rural areas
		CO2	village adoption for development
		CO3	Learn about education and health awareness programme
		CO4	Learn Social Responsibility
		CO5	NCC helps in Youth empowerment and helps in building of Nation
BAG-5201	**Fundamentals of genetics**	**CO1**	Apply the knowledge gained on inheritance and variation
		CO2	Study about structure and working of chromosomes
		CO3	Relate mutation to evolution and heredity
		CO4	Interpret the functions of genetic material.
BAG-5202	**Agricultural Microbiology**	**CO1**	Discriminate prokaryotic and eukaryotic microbes
		CO2	Delineate the structure and growth of bacteria
		CO3	Utilize microbes as models to study genetics
		CO4	Use microbes in enriching specific plant nutrients
		CO5	Analyze the ubiquitous nature of microbes inhabiting a wide range of ecological
BAG-5203	**Soil and Water Conservation Engineering**	**CO1**	Understand water and wind erosion and their mechanisms.
		CO2	Know various agronomical and mechanical measures for controlling soil erosion and moisture conservation.
		CO3	Develop analytical thinking and problem-solving skills in soil and water conservation engineering problems.
		CO4	Measure and estimate soil loss and sedimentation using different techniques.
		CO5	Design bunds, terraces, grassed waterways, wind breaks and shelter belts etc.
BAG-5204	**Fundamentals of Crop Physiology**	**CO1**	Understand different physiological process at plant and cellular level
		CO2	Understand mechanisms of uptake, transport and translocation of water and nutrients
		CO3	Understand carbon cycles in plants and define lipid metabolism
		CO4	Understand importance of growth regulators in plant growth
		CO5	Understand nutrient deficiencies and physiological requirements of plants
BAG-5205	**Fundamentals of Agricultural Economics**	**CO1**	Apply the knowledge gained on the fundamentals of economics
		CO2	Employ agricultural economic applications
		CO3	Practice applying mathematical models to agro-economics
		CO4	Interpret market structures responsible for creating national income
		CO5	Analyse agro-economic growth and develop policies
BAG-5206	**Fundamentals of Plant Pathology**	**CO1**	Importance and scope of plant pathology and analyse the causes and factors leading to pathogenesis
		CO2	Classify pathogens taxonomically for designing effective disease management strategies
		CO3	Understand plant pathogens based on morphology, vegetative, reproductive and resting structures.
		CO4	Relate disease cycles, physiology of pathogens and plant defence
		CO5	Understand epidemiology of plant diseases and strategies for disease management
BAG-5207	**Fundamentals of Entomology**	**CO1**	Understand the historic contributions of eminent scientists in the field of entomology and fascinating facts about insects

		CO2	Understand insect's anatomy and morphology
		CO3	Infer biochemical and physiological processes governing insect metabolism, growth, and form
		CO4	Relate ecological relationships of insects with other life forms
		CO5	Devise pest control measures
BAG-5208	**Fundamentals of Agricultural Extension Education**	**CO1**	Realize the necessity of agricultural extension for rural development
		CO2	Acquire knowledge on extension systems in India
		CO3	Devise plans for rural community development; plan and evaluate an extension programme
		CO4	Transfer technology and innovations towards agricultural development
		CO5	Develop interest in agricultural journalism
BAG-5209	**Communication Skills and Personality Development**	**CO1**	To have knowledge of proper use of grammar
		CO2	Knowledge of verbal and nonverbal way of expressing
		CO3	Way to have better presentation knowledge in presenting an article or in an interview and in group discussions
BAG-5301	**Crop Production Technology – I *(Kharif Crops)***	**CO1**	Identification of different Kharif crops.
		CO2	Familiar with different kharif crop varieties.
		CO3	Getting acquainted with Rice nursery preparations and precautions
		CO4	Getting knowledge of different sowing methods of kharif crops.
		CO5	Getting acquainted with basic fertilizers application methods, organic manures and bio-fertilizers.
BAG-5302	**Fundamentals of Plant Breeding**	**CO1**	Familiar with basics of Fundamentals of Plant breeding
		CO2	Familiar with different Plant Breeders kit.
		CO3	Getting acquainted with Emasculation & hybridization techniques.
		CO4	Study about the breeding methods for segregating generations field.
		CO5	Knowledge about Biotechnological tools.
BAG-5303	**Agricultural Finance and co-operation**	**CO1**	Familiar with Basics of Agriculture Finance & importance of Credit in Agriculture
		CO2	Acquainted with the knowledge concerning Sources of Agriculture Finance institutions as well as micro-finance schemes and about the functioning of higher financing institutions.
		CO3	Acquainted with the Balance Sheet (Assests& Liabilities) and Income Statement (Profit & loss account)
		CO4	Understanding the fundamentals of Techno-economic parameters for preparation of project report as well as SWOT analysis of an enterprise.
		CO5	Understand the significance of cooperatives in Indian agriculture
BAG-5304	**Agri-Informatics**	**CO1**	Familiar with Computer basics.
		CO2	Familiar with Internet & basic programming language.
		CO3	Knowledge of Use of Information Computer Technology in Agriculture
		CO4	Knowledge of IT application for Agri-input management
		CO5	Knowledge about Geospatial technology for generating valuable agri-information.
BAG-5305	**Farm Machinery and Power**	**CO1**	Familiar with Farm power & sources of farm power in India.

		CO2	Getting knowledge about Two Stroke and Four Stroke Engines, Working Principles, Applications- Types, Power and Efficiency
		CO3	Getting Familiar with different systems of I.C. engines
		CO4	Familiar with Power transmission systems
		CO5	Familiar with Primary & Secondary tillage implements concerning with farm
BAG-5306	**Production Technology for Vegetables and Spices**	**CO1**	Capable to identify different vegetables & spice crops and their seeds. Become aware about Importance of vegetables & spices in human nutrition.
		CO2	Getting knowledge about scientific production technology of different vegetables.
		CO3	Knowledge of morphological characters of different vegetables & spices.
		CO4	Getting acquainted with basic fertilizers application methods, organic manures and bio-fertilizers in Vegetables.
		CO5	Practical Knowledge regarding Harvesting & preparation for market.
BAG-5307	**Environmental Studies and Disaster Management**	**CO1**	Role of an individual in conservation of natural resources
		CO2	To aware about environmental protection acts
		CO3	To gain knowledge about the threats to biodiversity & its conservation principles.
		CO4	Awareness with respect to social development and welfare issues
		CO5	Meaning and nature of natural disasters, their types and effects and management
BAG-5308	**Statistical Methods**	**CO1**	Acquaintance with some basic concepts of statistics & its Applications in Agriculture
		CO2	Familiar with some elementary statistical methods like Probability , Binomial & Poisson Distributions
		CO3	Acquiring knowledge of Correlation, Scatter Diagram. Karl Pearson's Coefficient of Correlation
		CO4	Introduction to Test of Significance,
		CO5	Analysis of data pertaining to attributes and to interpret the results.
BAG-5309	**Livestock and Poultry Management**	**CO1**	Understand role of livestock in national economy, their housing and care management as well as management
		CO2	Understand different breeds of Livestocks , management of cattle's, buffalo, sheep , goat swine and poultry .Milking methods, Artificial insemination & their feed management
		CO3	Knowledge about Prevention (including vaccination schedule) and control of important diseases of livestock and poultry
BAG-5401	**Crop Production Technology –II (Rabi Crops)**	**CO1**	Knowledge about different Rabi crops
		CO2	Know various agronomical practices viz., crop variety, crop family, field preparation, seed treatment etc.
		CO3	Know about different sowing methods
		CO4	Identification of different growth stages of crops viz., tillering, branching, flowering etc.
		CO5	Knowledge about fertilizers and its application methods
BAG-5402	**Production Technology for Ornamental Crops, MAPs and Landscaping**	**CO1**	Identification of ornamental crops and their varieties
		CO2	Identification of medicinal and aromatic plants and their species
		CO3	Knowledge about cut flowers like rose, gerbera, carnation, lilium etc.

		CO4	Practice about bed preparation, seeds/ planting material treatments etc.
		CO5	Planning and layout of garden
BAG-5403	**Renewable Energy and Green Technology**	**CO1**	Understand the various forms of conventional energy resources.
		CO2	Understand concept of various forms of renewable energy
		CO3	Identification of solar cooker, solar water heater, solar drying, solar pond, solar pump, solar light, solar distillation etc.
		CO4	Compare Solar, Wind and bio energy systems, their prospects, Advantages and limitations.
		CO5	Understand the concept of Biomass energy resources and their classification, types of biogas Plants-applications
BAG-5404	**Problematic Soils and their Management**	**CO1**	Understand different kind of problem soil in India and there characteristics.
		CO2	Understand how to control or improve the soil fertility.
		CO3	Understand practical knowledge of laboratory to test the problem soil.
		CO4	Understand about Irrigation water and it quality.
BAG-5405	**Production Technology for Fruit and Plantation Crops**	**CO1**	Understand production techniques of tropical, sub-tropical, temperate fruit and plantation crops.
		CO2	Understand various propagation methods and important cultural practices for major fruit and plantation crops will be provided.
BAG-5406	**Principles of Seed Technology**	**CO1**	Understand about the various techniques of quality seed production, processing and seed quality enhancement
		CO2	Understand seed development, germination, vigor, deterioration and the relationship between laboratory tests and field performance.
		CO3	Understand seed increase systems, seed testing and the laws and regulations related to marketing high quality seed.
BAG-5407	**Farming System & Sustainable Agriculture**	**CO1**	Understand different cropping and farming system like integrated farming system (IFS).
		CO2	Understand sustainable agricultural practices such as organic farming.
		CO3	Understand cross-cultural interactions and exchange that linked the world’s people and facilitated agricultural development is also expected.
BAG-5408	**Agricultural Marketing Trade & Prices**	**CO1**	Understand Agricultural Marketing
		CO2	Understand of Demand, supply and producer’s surplus of agri-commodities
		CO3	Understand Product life cycle (PLC) and competitive strategies
		CO4	Understand Marketing process and functions
		CO5	Understand market integration; marketing efficiency; marketing costs, margins and price spread
BAG-5409	**Introductory Agro-meteorology & Climate Change**	**CO1**	Understand agro meteorology
		CO2	Understand roles of agro meteorology in agriculture and its relation to other areas of agriculture.
		CO3	Understand different metrological parameters like rainfall, temperature, RH and other weather parameters.
		CO4	Understand characteristics, behaviour or phenomenon of the atmosphere.
		CO5	Understand changes of individual weather elements
BAG-5410	**Weed Management**	**CO1**	Understand the place of weeds in agriculture.

	(Elective Course)	**CO2**	Understand weeds based on different characteristics
		CO3	Understand Planning for their control, control options, surveillance, herbicide trials
		CO4	Understand about control methods and use of herbicides
		CO5	Understand Approaches to weed management
BAG-5501	**Principles of Integrated Pest and Disease Management**	**CO1**	To know Insect Vectors transmitting plant diseases.
		CO2	Knowledge of Insect Control Methods.
		CO3	To know about Plant Protection equipment's.
BAG-5502	**Manures, Fertilizers and Soil Fertility Management**	**CO1**	To impart knowledge on soil essential nutrients and nutrient transformations in soil .
		CO2	To know the soil fertility management.
		CO3	The knowledge gained by students through this course will be useful in making decisions on nutrient dose, choice of fertilizers/manures and method of application etc.
		CO4	The students will also gain confidence in managing soil health for sustained productivity.
BAG-5503	**Pests of Crops and Stored Grains and their Management**	**CO1**	Students will know about pest of crops and stored grains like cereals, pulses, oilseeds and their management.
		CO2	Comprehend grain store management.
		CO3	Assess losses created due to insect pests in crops and recommend control measures
BAG-5504	**Diseases of Field and Horticultural Crops and their Management -I**	**CO1**	Understand diseases of various Field crops and Horticultural crops and to know their management practices
BAG-5505	**Crop Improvement-I (Kharif Crops)**	**CO1**	Understand different methods of plant breeding like pure line selection, mass selection, pedigree method and other hybrid crop varieties production for special crop improvement.
		CO2	Understand Crop improvement may be for drought resistance, high yield, pest and disease resistance.
BAG-5506	**Entrepreneurship Development and Business Communication**	**CO1**	Understand Concept of Entrepreneur, Entrepreneurship Development
		CO2	SWOT Analysis & achievement motivation, Government policy and programs
BAG-5507	**Geo-informatics and Nano-technology and Precision Farming**	**CO1**	Understandapplications of GIS in agriculture which will help them to forecast for precision farming.
BAG-5508	**Practical Crop Production-I *(Kharif Crops)***	**CO1**	To do all field operations in the allotted land from field preparation to harvest and processing.
		CO2	Understand water scarcity to raise wetland rice of the crop production programme
		CO3	Irrigated puddled lowland rice will be cultivated.
BAG-5509	**Intellectual Property Rights**	**CO1**	Students will be aware of Intellectual Property Rights for ensuring rights for their products.
BAG-5510	**Agrochemicals(Elective Course)**	**CO1**	Understanding the role of agrochemicals in agriculture and its effect on environment
		CO2	Imparting knowledge on herbicides, fungicides, insecticides, fertilizers and its applications
		CO3	Emphasising the use of right dose of agrochemicals for sustainable agriculture
BAG-5601	**Rainfed Agriculture and Watershed Management**	**CO1**	Basic knowledge of rain fed agriculture and water shed management
		CO2	Understand objective, principles and component of watershed management
		CO3	Student can able to understand about rainfed agriculture and its introduction, problem and prospects in India

BAG-5602	**Protected Cultivation and Secondary Agriculture**	**CO1**	Understand Green house technology
		CO2	Understand Important of Protected Cultivation
		CO3	Understand how to grow plant in protected condition
BAG-5603	**Diseases of Field and Horticultural Crops and their Management-II**	**CO1**	Understand different type of pathogens occurs in horticultural crops & Field crops.
		CO2	Understand diseases of the field and horticultural crops.
		CO3	Understand diseases suitable management methods can be applied.
BAG-5604	**Post-harvest Management and Value Addition of Fruits and Vegetables**	**CO1**	Understand importance of post harvest management
		CO2	Understand methods and process of preservation
		CO3	Understand how to manage fruits and vegetables after harvesting
BAG-5605	**Management of Beneficial Insects**	**CO1**	Understand beneficial insects with respect to its commercial use in agriculture.
		CO2	Understand rearing of beneficial insects commercially along with its use in pest control
BAG-5606	**Crop Improvement – II (*Rabi*)**	**CO1**	Understand knowledge of rabi crops and it`s crop improvement approach
		CO2	Understand major plant breeding approach of rabi crops
		CO3	Understand hybrid seed production technology of rabi crops
BAG-5607	**Practical Crop Production –II (Rabi crops)**	**CO1**	Understand package and practices of Rabi crops
		CO2	Understand the preparation field for rising crop
		CO3	Understand the package and practices of Rabi crops
BAG-5608	**Principles of Organic Farming**	**CO1**	Understand how to produce organic product
		CO2	Understand the importance and principles of organic farming
		CO3	Understand Certification process and standards of organic farming
BAG-5609	**Farm Management, Production & Resource Economics**	**CO1**	Understand principles applied Farm Management dealing with the analysis of limited farm resources to students.
		CO2	Understand different types of farms and economic principles applied to manage farms
		CO3	Understand to prepare budgeting of farms as well as different enterprises of farms.
		CO4	Understand resource management strategy to achieve economic and sustainable production of farms.
BAG-5610	**Principles of Food Science and Nutrition**	**CO1**	Understand differencing redients of food and their chemistry
		CO2	Understand principles and methods of preservation and processing of food
		CO3	Understand roles of different microbes in food items.
		CO4	Understand correlation between food, nutrition and manner to overcome malnutrition problems.
BAG-5611	**Bio pesticides&Bio fertilizers (Elective Course)**	**CO1**	Understand about bio-pesticides and bio fertilizers its uses and utility in crop husbandry.
		CO2	Understand awareness of bio-pesticides and bio fertilizers. Methods of preparation and application.
		CO3	Understand importance of bio pesticides and bio fertilizers over commercial chemical pesticides and fertilizers.

B.Tech (Agricultural Engineering)

Program Outcomes

PO 1 **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO 2 **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO 3 **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO 4 **Conduct investigations of complex problems**: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO 5 **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO 6 **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO 7 **Environment and sustainability**: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO 8 **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO 9 **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO 10 **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO 11 **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO 12 **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change

Program Specific Outcomes

At the end of the programme, the Agricultural Engineering graduates will be able to

PSO 1: Identify and design sustainable solutions to solve agricultural engineering problems applying the knowledge of basic and engineering sciences.

PSO 2: Pursue successful professional career in agro-industries, government organization, educational/ research/ extension institute adopting appropriate technology, resources and modelling.

PSO 3: Adapt in a world of evolving technology with professional ethics & take lead roles in development of agricultural engineering & allied enterprises for betterment of society.

Course Outcomes

BTA-5101	**Engineering Mathematics-I**	**CO1**	Knowledge about matrices and their ranking and Understand the Eigen values, Eigen vectors and their properties
		CO2	Develop knowledge on Cayley Hamilton theorem And Knowledge about diagonalization of matrices
		CO3	Understanding the function of two or more independent variables and Distinguish between homogenous and composite function
		CO4	Knowledge about Taylor's and Maclaurin's expansion
		CO5	Have a brief knowledge on Jacobian's error evaluation and Know about Stroke's divergence and Green's theorem.
BTA-5102	**Engineering Physics**	**CO1**	Brief knowledge on conductor, insulator and semiconductors
		CO2	Knowledge about the quantum nature of electrons and effective mass of electrons and holes & understand dual nature of particles
		CO3	Knowledge about different types of magnetic materials and their properties and Understand the effect of electric and magnetic field on spectra of atoms and molecules
		CO4	Understanding of holography and knowledge about super conducting material and their properties
		CO5	Knowledge about optical fibres and its application and illumination and its laws.
BTA-5103	**Engineering Chemistry**	**CO1**	Knowledge about different type of fuel and their quality through various analytical methods
		CO2	Understand the cause, effect and prevention of corrosion
		CO3	Knowledge about the nutrition value of food, types of food preservatives, colouring reagents and flavouring reagents
		CO4	Acquire basic idea about polymerization and its properties and determination of molecular mass through various methods
		CO5	Learn about the type of water and its effect on industries and Acquire knowledge about the modern instruments for analysis of samples
BTA-5104	**Principles of Soil Science**	**CO1**	Understand the origin of soil, soil forming rocks minerals, their classification and composition.

		CO2	Know about the physical and chemical properties of soil.
		CO3	Acquaint with the important of clay minerals & their role in nutrient transformation in soil
		CO4	Gain knowledge regarding the role of plant nutrients, deficiency symptoms, important fertilizer and their reaction in soil.
		CO5	Know about the problematic soil, their characteristics and management.
BTA-5105	**Surveying and Leveling**	**CO1**	Understand the importance of surveying, how to prepare maps, plans and the importance of different scales
		CO2	Know about taking measurements linear, angular and altitude etc.
		CO3	Know about the different land features, specifications of work and to give layout of Engineering structures
		CO4	Computation of area of fields and volume of earthwork, items of works in construction
BTA-5106	**Engineering Mechanics**	**CO1**	Understand the importance of different force system, composition, resolution and analysis of force systems
		CO2	Know about moments, couples, C.G., moment of inertia, FBD and frictional forces
		CO3	Know about the different simple framed structures, shear force, stresses and torsion in the beams
BTA-5107	**Engineering Drawing**	**CO1**	Know the principle & construction of different types of scales, geometrical figures and understand the principle of isometric projection and able to draw the same
		CO2	Able to draw 1^{st}& 3^{rd}angle projection of points , lines and solids
		CO3	Understand the principle of development of surfaces, section of solids and able to draw the same
		CO4	Understand the principle of conversion of pictorial view to orthographic view and able to draw the same
		CO5	Drawing of screw thread, nuts and bolts, welding symbol, rivet joint etc
BTA-5108	**Heat and Mass Transfer**	**CO1**	Understand the principles of conduction, convection, radiation
		CO2	Analyse the mathematical and practical aspect of conduction, convection and radiation
		CO3	Analyse the mathematical and practical aspect of heat exchangers
		CO4	Understand the principles of mass transfer and related laws.
BTA-5201	**Engineering Mathematics–II**	**CO1**	Understand Cauchy’s and Legendre’s linear equations and knowledge about series solution techniques
		CO2	Distinguish between the exact and Bernoulli’s differential equations and apply of one dimensional heat and wave equation
		CO3	Distinguish between various functions of complex variables and understand cauchyreimann equation
		CO4	Knowledge about ratio and root test and formation of partial differential equation
		CO5	Knowledge about lagrange’s linear equation,charpit’s method and even and odd functions and their fourier series
BTA-5202	**Environmental Science and Disaster Management**	**CO1**	Understand different resources such as mineral resources, food resources, water resource, energy resources, natural resources and land resources.
		CO2	Gather knowledge about environmental pollution, Soil

			pollution, Air pollution and thermal pollution
		CO3	Acquaint themselves with waste land reclamation, Ecosystems and their management
		CO4	Learn regarding Biodiversity and its conservation, natural disaster and manmade disaster along with their management
BTA-5203	**Entrepreneurship Development & Business Management**	**CO1**	Understanding the basic concept of entrepreneur, entrepreneurship development process and skill required becoming an entrepreneur.
		CO2	Analysis of internal and external environment for business opportunity and government policy available for entrepreneurs
		CO3	Understanding business communication skill, managerial skill and problem solving skill required to become an entrepreneur
		CO4	Learning of project preparation techniques
		CO5	Learn about process of set up new enterprises
BTA-5204	**Fluid Mechanics & Open Channel Hydraulics**	**CO1**	Understand the properties of fluid
		CO2	Knowledge of the behaviour of the fluid at rest
		CO3	Knowledge of the behaviour of the fluid in motion
		CO4	Application of the principles of fluid mechanics to design simple fluid mechanical systems in engineering
BTA-5205	**Strength of Materials**	**CO1**	Understand the importance of strength parameters
		CO2	Calculate deflection, bending moment and shear force to assess the stress conditions
		CO3	Know the techniques to calculate unknown forces in 2D structures.
BTA-5206	**Workshop Technology and Practice**	**CO1**	Understanding the importance of workshop technology.
		CO2	Learning about analyse of applications of different shops.
		CO3	Knowledge about the basic manufacturing process involved for production of different machine elements
BTA-5207	**Theory of Machines**	**CO1**	Understand the importance of theory behind the machine &structure
		CO2	Analyse the relative motion between the various parts of machine, and forces which act on them.
		CO3	The knowledge of this subject is very essential for an engineer in designing the various parts of the machine.
		CO4	Recommend methods for balancing a component.
BTA-5208	**Web Designing and Internet Applications**	**CO1**	understanding the internet and its necessity
		CO2	knowledge of the web design principles
		CO3	Knowledge about using java script how to make interactive webpages.
BTA-5301	**Principles of Horticultural Crops and Plant Protection**	**CO1**	identification of different horticultural crops and knowledge of methods of macro and micro-propagation
		CO2	getting acquainted with different types of basic fertilizers, organic manures & bio-fertilizers and understanding of application of fertilizer and irrigation water to horticultural crops
		CO3	knowledge about training and pruning methods, maturity time for harvesting of crop and knowledge of the vegetable seed extraction technique
		CO4	learning about importance of nursery raising, basics of preparation of potting mixture, potting and repotting of plants and identification and use the garden tools
		CO5	identification about different major pests and diseases of horticultural crops
BTA-5302	**Principles of Agronomy**	**CO1**	Develop knowledge of the crop types, variety, seasons

			of sowing, sowing/transplanting methods, crop development stages & the effect of weather parameters on crop production
		CO2	Develop knowledge on different tillage practices in the crop field
		CO3	Learn the time and methods of application of different types of Fertilizers & Organic Manures
		CO4	understanding soil water constants and their role in crop production & weeds and know the methods of weed control
		CO5	knowledge about various cropping patterns and their application
BTA-5303	**Communication Skills and Personality Development**	**CO1**	Develop academic and social English competencies in speaking,
		CO2	Listening and pronunciation, develop reading and writing skills
		CO3	Learn grammar and usage, vocabulary, syntax, and rhetorical patterns.
BTA-5304	**Engineering Mathematics III**	**CO1**	knowledge about finite difference methods, operators and their relations
		CO2	understanding the newton’s forward, backward, beselles’s, stirling’s central difference interpolations and lagranges interpolation formula
		CO3	knowledge about numerical integration by trapezoidal and simpson’s rule and understanding of rules for finding complimentary function and particular integral
		CO4	Know about properties of Laplace transform, inverse of Laplace transform and their applications
		CO5	Application of Laplace’s transform to solve differential and simultaneous differential equations
BTA-5305	**Soil Mechanics**	**CO1**	knowledge of different engineering properties like moisture content, density, void ratio, porosity, grain size analysis etc.
		CO2	Knowledge of about different soil consistencies like liquid limit, plastic limit, and shrinkage limit.
		CO3	knowledge about different stress conditions like shear stress, direct stress and stress due to different loading conditions
		CO4	knowledge about consolidation and compaction properties of soil
		CO5	learn about different active and passive earth pressure on retaining walls and know about different stability conditions of slopes
BTA-5306	**Design of Structures**	**CO1**	Understand the importance of structural design
		CO2	Calculate loads on a structure
		CO3	Design small and medium RCC and steel structures
BTA-5307	**Machine Design**	**CO1**	Understand the designer’s work concerned with adaption of existing design.
		CO2	To modify the existing design into new idea by adopting a new material or different method of manufacturing.
		CO3	May take up the work of new design.
BTA-5308	**Thermodynamics, Refrigeration and Air Conditioning**	**CO1**	Understand the basics of thermodynamics.
		CO2	Understand the principle of various refrigeration cycles.
		CO3	Understand the principle of psychometric processes and air conditioning.
		CO4	Design of air conditioning system.
BTA-5309	**Electrical Machines and Power Utilization**	**CO1**	Different types of circuits and their applications.
		CO2	Principles and operation of transformers, DC machines

	Conditioning		and motors.
		CO3	Various methods of power measurement
BTA-5401	**Building Construction and Cost Estimation**	**CO1**	Understand the importance of various building materials for construction work
		CO2	Know about various components of a building with items of work
		CO3	Know about methods of construction of agricultural buildings, sloped and flat roofed buildings
		CO4	Know about preparation of various types of estimates of buildings
		CO5	To take measurements, find depreciation, cost-in-use, benefit-to-cost, net benefits, payback period etc.
BTA-5402	**Auto CAD Applications**	**CO1**	To introduce to the students the importance of machine drawing in engineering applications.
		CO2	To impart the knowledge of assembly drawing and production drawing of machineComponents.
		CO3	The Computer Graphics course will enable the student to gain knowledge on how computers are integrated at various levels of planning and manufacturing
		CO4	To understand the flexible manufacturing system and to handle the product data and various software used for manufacturing.
BTA-5403	**Applied Electronics and Instrumentation**	**CO1**	Knowledge of the operating principle of various electronics device and the use of different measuring instrument in different sector.
		CO2	Learn about characteristics of diode and transistor.
		CO3	Application of opamp in various purposes.
		CO4	Application of diode as clipper and clamper circuit.
		CO5	Learn about various types of transducers.
BTA-5404	**Tractor and Automotive Engines**	**CO1**	Understand the theory of engines, their types and usefulness
		CO2	Understand the working principle of different systems of internal combustion engines and tractor.
		CO3	To study the working principle, maintenance and adjustment of different engine systems
BTA-5405	**Engineering Properties of Agricultural Produce**	**CO1**	Understand the various engineering properties like physical, thermal, frictional, aerodynamic, rheological and electrical properties of agricultural produce.
		CO2	Know the methods of measurement physical, thermal, frictional properties of granular Material
		CO3	Knowledge of Rheological and Electrical properties of Biological material
		CO4	Understand the importance of these properties in handling processing machines and storage structures
BTA-5406	**Watershed Hydrology**	**CO1**	Understand the relevance of various components of hydrologic cycle, which are responsible for spatial and temporal distribution of water availability in any region.
		CO2	Quantify different hydrological processes, know their methods of analysis for application in hydrological studies.
		CO3	Apply ideas and techniques in related areas of watershed development, water harvesting, minor irrigation, drought and flood control etc.
BTA-5407	**Irrigation Engineering**	**CO1**	Knowledge of methods to measure the soil moisture content in the field & knowledge of procedures to measure discharge of water flowing in the irrigation channel.
		CO2	Understand the methodologies to compute the water requirement of crops. And Knowledge of how to

			design the underground water conveyance system and irrigation channels
		CO3	Knowledge of lining materials used to reduce the seepage loss in channels and Measurement the bulk density, field capacity and wilting point.
		CO4	Knowledge of different irrigation methods like: furrow, border, check basin. Calculation of irrigation efficiency.
		CO5	Be acquainted with accurate irrigation scheduling with proper methods of irrigation for different crops.
BTA-5408	**Sprinkler and Micro irrigation Systems**	**CO1**	To understand the importance of micro-irrigation (drip and sprinkler, role of Government for promotion
		CO2	To acquaint the students about the components of micro irrigation systems, their design and lay out for efficient water, fertilizer and pesticides application
		CO3	To gain knowledge about the hydraulics of sprinkler and drip irrigation
		CO4	Benefit – cost analysis in micro irrigation systems
BTA-5409	**Fundamentals of Renewable Energy Sources**	**CO1**	To harness the environment friendly RE sources and to enhance their contribution to the socio economic development.
		CO2	To meet and supplement rural energy needs through sustainable RE projects.
		CO3	To provide decentralized energy supply to agriculture, industry, commercial and household sector.
		CO4	On successful completion of the teaching program, the students should be able to evaluate, and select appropriate energy technologies to meet a given energy demand.
BTA-5501	**Tractor Systems and Controls**	**CO1**	Knowledge about the importance of different systems in a tractor.
		CO2	Knowledge about transmission and brake system
		CO3	Knowledge about the steering and hydraulic system
		CO4	Knowledge about the power outlets and tractor chassis mechanics
		CO5	Knowledge about safety consideration in tractor and standards of tractor testing.
BTA-5502	**Farm Machinery and Equipment-I**	**CO1**	Know about the farm machineries used in agricultural production and construction, operation of different machines
		CO2	Know about the operating parameters and performance of the machines
		CO3	Know about the cost of operations and economics of the machines.
		CO4	Study the different components of the machines, their constructions and Materials of construction.
		CO5	Knowledge about adjustments of different components to enhance performance.
BTA-5503	**Agricultural Structures &Environmental Control**	**CO1**	Understand the planning and lay out of a farmstead
		CO2	Know about various standards for various dairy, piggery, poultry and other farm structures.
		CO3	Know about the different farm storage structures, silos, compost pit, implement sheds, farm houses, threshing floors, farm roads, fencing, water supply, sewage systems and septic tanks
		CO4	Know about rural electrification, concepts of eco system, bio-diversity, environmental pollution and control, solid waste, plant waste management
		CO5	To prepare estimate for different farm buildings, structures, roads, fencing and construction, repair and

			maintenance of farm structures.
BTA-5504	**Post-Harvest Engineering of Cereals, Pulses & Oil Seeds**	**CO1**	Know the different unit operations in processing of major cereals, pulses &oilseeds of the country & state
		CO2	Understand the working principles of different machineries used for processing of cereals, pulses and oilseeds
		CO3	The basics of selection of appropriate machines/ equipment for various applications
		CO4	Know the different uses of by-products obtained from cereals, pulses and oilseeds Processing.
BTA-5505	**Soil and Water Conservation Engineering**	**CO1**	Understand water and wind erosion and their mechanisms.
		CO2	Know various agronomical and mechanical measures for controlling soil erosion and moisture conservation.
		CO3	Develop analytical thinking and problem solving skills in soil and water conservation engineering problems.
		CO4	Measure and estimate soil loss and sedimentation using different techniques.
		CO5	Design bunds, terraces, grassed waterways, wind breaks and shelter belts etc. and recommend them for better soil and moisture conservation in a watershed.
BTA-5506	**Watershed Planning and Management**	**CO1**	Knowledge about the problems of watershed, and characteristics
		CO2	Knowledge about different concept and factors affecting on watershed.
		CO3	Knowledge about Rainwater conservation technologies
		CO4	Knowledge about Monitoring and evaluation of a watershed along with PRA study and cost analysis
		CO5	Understanding of delineate a watershed.
BTA-5507	**Drainage Engineering**	**CO1**	Understand the drainage problems and their effect on crop production
		CO2	Know about the salt affected soils and their reclamation methods
		CO3	Distinguish between surface, sub-surface and non-conventional drainage methods
		CO4	Investigation methods of drainage problems and design drain dimensions & the spacing between drains
		CO5	Identify the drainage materials and study their strength prior to installation and Evolve methods to design the surface and sub-surface drainage systems
BTA-5508	**Renewable Power Sources**	**CO1**	To understand the different energy consumption pattern and replace of renewable energy
		CO2	Learning about design of community type and institutional type of biogas plants and its use for IC engine operation for power generation
		CO3	Understand solar thermal, photovoltaic power generation and detail design of wind mill and its power operation
		CO4	Understand ocean power generation through OTEC, wave and tide and power generation from urban waste and biomass gasification
		CO5	Learning about design of mini and small hydro project, detail study of fuel cell and their application and use of animal energy for different agricultural operation
BTA-5601	**Computer Programming and Data Structures**	**CO1**	Understand the high level language.
		CO2	Understand the algorithm of a problem and coding of that algorithm.
		CO3	Know all the details of data types, operators, library functions, decision making, branching, looping,

			arrays, user defined functions, recursion, string functions, structures and union, pointers, stacks, queues, linked lists.
BTA-5602	**Farm Machinery and Equipment-II**	**CO1**	Know about the farm machineries used in agricultural production
		CO2	Know about construction, operation of different machines
		CO3	Know about the operating parameters and performance of the machines
		CO4	Know about the cost of operations and economics of the machines
		CO5	To solve numerical on different chapters
BTA-5603	**Post-Harvest Engineering of Horticultural Crops**	**CO1**	Know the different means of storage and value addition of fruits and vegetables along with the cold chain
		CO2	Know the different unit operations in processing of major horticultural crops of the country and state
		CO3	Understand the working principles of different machineries used for processing of fruits, vegetables and spices
		CO4	Understand the basics of selection of appropriate machines/ equipment for various applications of processing of horticultural crops
BTA-5604	**Water Harvesting and Soil Conservation Structures**	**CO1**	Know different soil erosion control structures and their requirements for better soil and moisture conservation in different situations.
		CO2	Understand the theory behind flow through soil conservation structures and use specific energy and momentum concepts to analyze flow problems.
		CO3	Prepare plan and design water harvesting and permanent soil and water conservation engineering structures with cost estimation.
		CO4	Design permanent soil and water conservation structures like drop spillway, chute spillway, drop inlet spillways, earthen embankments and water harvesting structures and estimate their cost.
		CO5	Recommend different structures for better soil and moisture conservation in problem areas.
BTA-5605	**Groundwater, Wells and Pumps**	**CO1**	To understand the knowledge on occurrence,distribution and hydraulics of ground water in a region/area and characteristics/properties of the water bearing formations.
		CO2	To acquaint with the equipment & different methods used for construction of wells suitable for different formations and the design of deep/ shallow wells.
		CO3	To have knowledge on different recharge methods for augmentation of ground water. know the quality of ground water and its suitability for irrigation, drinking and industry purpose.
		CO4	To acquaint with different types of water lifting devices such as pumps, hydraulic ram, their adaptability and selection of required size.
		CO5	Selection of site for construction of wells. Efficient design of wells in the field condition.
BTA-5606	**Tractor and Farm Machinery Operation and Maintenance**	**CO1**	To enable the students to know the operation of tractor and farm machinery.
		CO2	To enable the students to understand the maintenance of tractor and farm machinery.
BTA-5607	**Dairy and Food Engineering**	**CO1**	Understand the different unit operations in processing and value addition of different dairy and food

			products.
		CO2	Understand the different types of equipment and their working principles used for processing and dairy and food products.
		CO3	Understand the basic principles of selection, operation and maintenance of different equipment for processing of milk and other food products for storage and value addition.
		CO4	Guide a prospective entrepreneur for basic layout and selection of equipment for a food processing plant.
BTA-5608	**Bio-Energy Systems: Design and Applications**	**CO1**	The different biomass sources, their properties and their applications for energy
		CO2	Production of biomass
		CO3	Thermo chemical and biochemical process of biomass for bioenergy and fuel production
BTA-5801	**Development of Processed Products**	**CO1**	Unit operations and equipment used for different food processing operations.
		CO2	Processing technologies for value addition of cereals, pulses, oilseeds, vegetables, fruits, milk, fish, meat and poultry products.
		CO3	Know the operations and equipment used in food processing industries
		CO4	Have a basic understanding of process technologies for processing of cereals, pulses, oilseeds, vegetables, fruits, milk, fish, meat and poultry.
BTA-5802	**Tractor Design and Testing**	**CO1**	Procedure for design and development of agricultural tractor, and Parameters for balanced design of tractor for stability & weight distribution,
		CO2	Understanding the Traction theory, hydraulic lift and hitch system design. And Design of mechanical power transmission in agricultural tractors: single disc, multi disc and cone clutches
		CO3	Learn about Rolling friction and anti-friction bearings. And Design of Ackerman Steering and tractor hydraulic steering.
		CO4	Study of special design features of tractor engines and their selection viz. cylinder, piston, piston pin, crankshaft, etc.
		CO5	Learn about Design of seat and controls of an agricultural tractor. Tractor Testing.
BTA-5803	**Wasteland Development**	**CO1**	To understand the concept of land degradation, classification and mapping of wastelands.
		CO2	To acquaint the students about planning of the wasteland development keeping in view agro climatic conditions, development options, contingency plans, conservation measures, water harvesting and recycling methods in consideration.
		CO3	To gain knowledge on different land reclamation and rehabilitation measures for wasteland development.
		CO4	To impart knowledge in the use of micro-irrigation for sustainable wasteland development against adverse situations like drought and water source situations.